36-315 Project

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Final Project [Change title]

Motivation section

What is the quality of these elements? Does the title tell us what to expect of the report to follow? Does the motivation section convince the reader that the broader study of this data set could elicit interesting insights?

Description of Dataset

```r
library(tidyverse)
library(ggplot2)

shopping_data <- read.csv("shopping_behavior_updated.csv")
head(shopping_data)
```

```
##   Customer.ID Age Gender Item.Purchased Category Purchase.Amount..USD.
## 1           1  55   Male         Blouse Clothing                    53
## 2           2  19   Male        Sweater Clothing                    64
## 3           3  50   Male          Jeans Clothing                    73
## 4           4  21   Male        Sandals Footwear                    90
## 5           5  45   Male         Blouse Clothing                    49
## 6           6  46   Male       Sneakers Footwear                    20
##        Location Size     Color Season Review.Rating Subscription.Status
## 1      Kentucky    L      Gray Winter           3.1                 Yes
## 2         Maine    L    Maroon Winter           3.1                 Yes
## 3 Massachusetts    S    Maroon Spring           3.1                 Yes
## 4  Rhode Island    M    Maroon Spring           3.5                 Yes
## 5        Oregon    M Turquoise Spring           2.7                 Yes
## 6       Wyoming    M     White Summer           2.9                 Yes
##   Shipping.Type Discount.Applied Promo.Code.Used Previous.Purchases
## 1       Express              Yes             Yes                 14
## 2       Express              Yes             Yes                  2
## 3 Free Shipping              Yes             Yes                 23
## 4  Next Day Air              Yes             Yes                 49
## 5 Free Shipping              Yes             Yes                 31
## 6      Standard              Yes             Yes                 14
##   Payment.Method Frequency.of.Purchases
## 1          Venmo            Fortnightly
## 2           Cash            Fortnightly
## 3    Credit Card                 Weekly
## 4         PayPal                 Weekly
## 5         PayPal               Annually
## 6          Venmo                 Weekly
```

What is the quality of this description? Does it clearly communicate what the rows and columns (i.e., observations and variables) are in the dataset in a way that is understandable to a CMU statistics undergraduate audience?

3 Research Questions

What is the quality of the research questions? Are they well-motivated by real-world/scientific interests? Or are they shallow? For example, a very shallow question would be in the form, "what does the distribution of this variable look like?" - three questions like this would guarantee 0pts for this item. A more interesting question would motivate why we would like to inspect particular distributions with real-world context

Question 1

Graph, Description

Question 2

Graph, Description

Question 3

Graph, Description

Evaluation/Future work

Old Draft

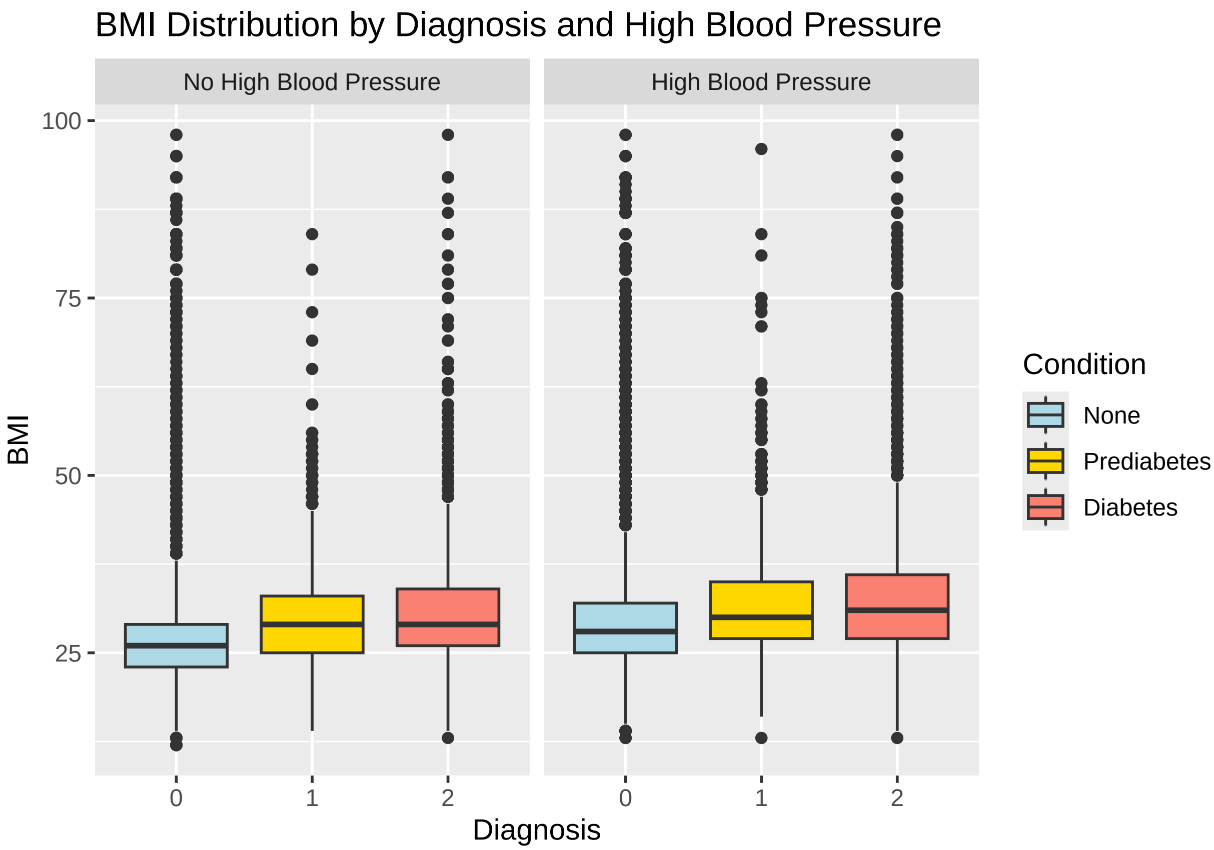
```r
library(tidyverse)
library(ggplot2)

url <- "https://raw.githubusercontent.com/AliceeLe/36315/main/diabetes_012_health_indicators_BRFSS2015.c
diabetes_dataset <- read.csv(url, header = TRUE, stringsAsFactors = FALSE)

head(diabetes_dataset)
```

```
##   Diabetes_012 HighBP HighChol CholCheck BMI Smoker Stroke HeartDiseaseorAttack
## 1            0      1        1         1  40      1      0                    0
## 2            0      0        0         0  25      1      0                    0
## 3            0      1        1         1  28      0      0                    0
## 4            0      1        0         1  27      0      0                    0
## 5            0      1        1         1  24      0      0                    0
## 6            0      1        1         1  25      1      0                    0
##   PhysActivity Fruits Veggies HvyAlcoholConsump AnyHealthcare NoDocbcCost
## 1            0      0       1                 0             1           0
## 2            1      0       0                 0             0           1
## 3            0      1       0                 0             1           1
## 4            1      1       1                 0             1           0
## 5            1      1       1                 0             1           0
## 6            1      1       1                 0             1           0
##   GenHlth MentHlth PhysHlth DiffWalk Sex Age Education Income
## 1       5       18       15        1   0   9         4      3
## 2       3        0        0        0   0   7         6      1
## 3       5       30       30        1   0   9         4      8
## 4       2        0        0        0   0  11         3      6
## 5       2        3        0        0   0  11         5      4
## 6       2        0        2        0   1  10         6      8
```

EDA

Alice's EDA

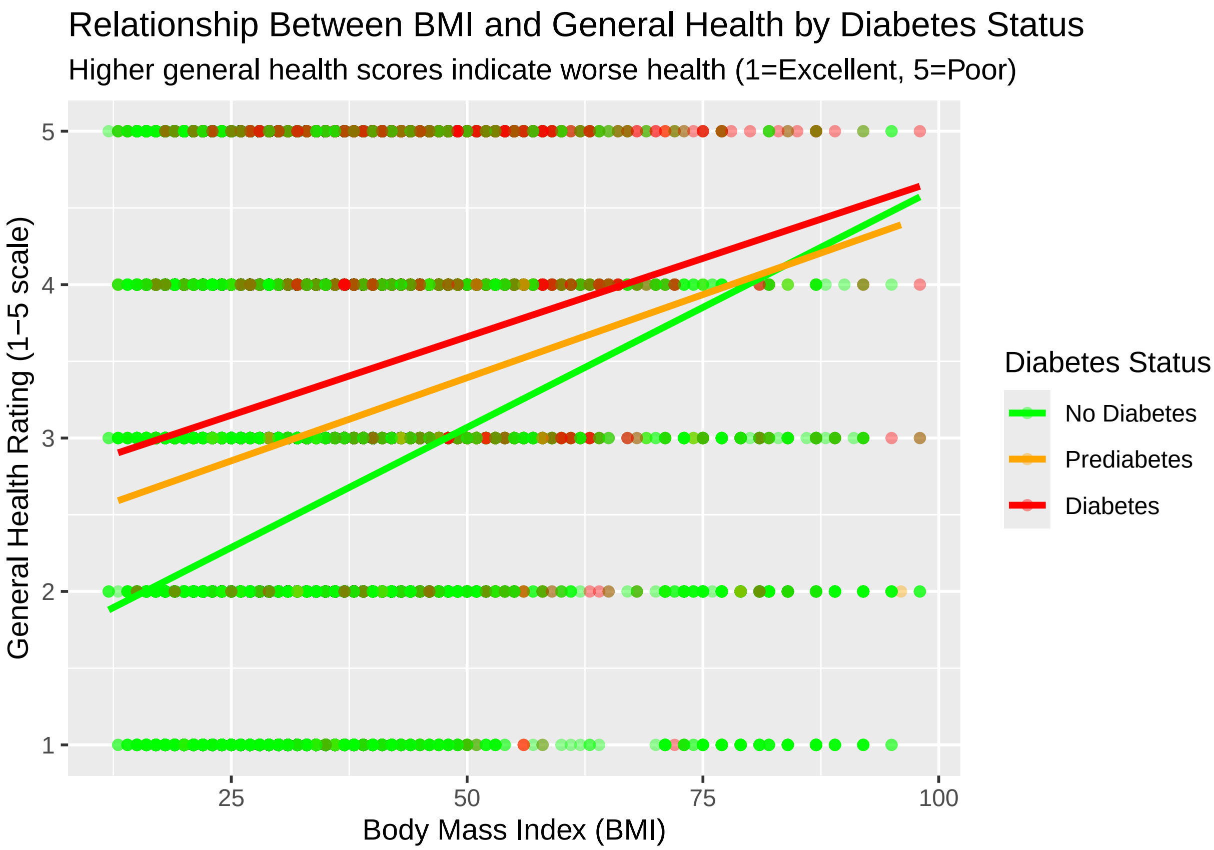
```r
ggplot(diabetes_dataset, aes(x = factor(Diabetes_012), y = BMI, fill = factor(Diabetes_012))) +
  geom_boxplot() +
  facet_wrap(~ HighBP,
             labeller = labeller(HighBP = c("0" = "No High Blood Pressure", "1" = "High Blood Pressure")
  scale_fill_manual(values = c("0" = "lightblue", "1" = "gold", "2" = "salmon"),
                    labels = c("0" = "None", "1" = "Prediabetes", "2" = "Diabetes")) +
  labs(x = "Diagnosis",
       y = "BMI",
       fill = "Condition",
       title = "BMI Distribution by Diagnosis and High Blood Pressure")
```



Hailey's EDA

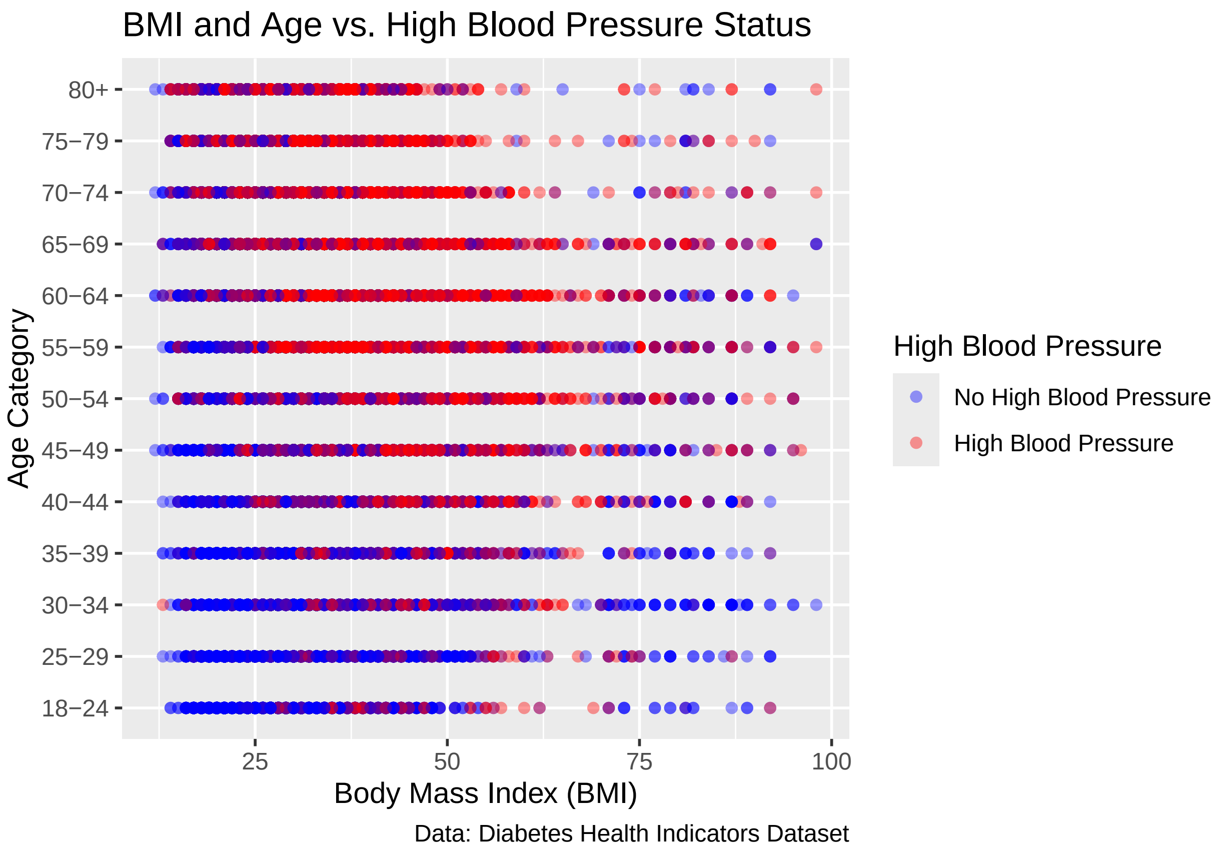
```
diabetes_dataset %>%
  mutate(
    Diabetes_Status = factor(Diabetes_012,
                             levels = c(0, 1, 2),
                             labels = c("No Diabetes", "Prediabetes", "Diabetes")),
    High_BP = factor(HighBP,
                     levels = c(0, 1),
                     labels = c("Normal BP", "High BP"))
  ) %>%
  ggplot(aes(x = BMI, y = GenHlth, color = Diabetes_Status)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.2) +
  scale_color_manual(values = c("No Diabetes" = "green",
                                "Prediabetes" = "orange",
                                "Diabetes" = "red")) +
  labs(
    title = "Relationship Between BMI and General Health by Diabetes Status",
    subtitle = "Higher general health scores indicate worse health (1=Excellent, 5=Poor)",
    x = "Body Mass Index (BMI)",
    y = "General Health Rating (1-5 scale)",
    color = "Diabetes Status"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Michelle's EDA

```
diabetesByAge <- diabetes_dataset %>%
  mutate(AgeCategory = factor(Age, levels = sort(unique(Age)),
                              labels = c("18-24","25-29","30-34","35-39","40-44","45-49",
                                         "50-54","55-59","60-64","65-69","70-74","75-79","80+")))
ggplot(diabetesByAge, aes(x = BMI,
                 y = AgeCategory,
                 color = factor(HighBP))) +
  geom_point(alpha = 0.4) +
  scale_color_manual(values = c("blue", "red"),
                     labels = c("No High Blood Pressure", "High Blood Pressure")) +
  labs(
    title = "BMI and Age vs. High Blood Pressure Status",
    x = "Body Mass Index (BMI)",
    y = "Age Category",
    color = "High Blood Pressure",
    caption = "Data: Diabetes Health Indicators Dataset"
  )
```

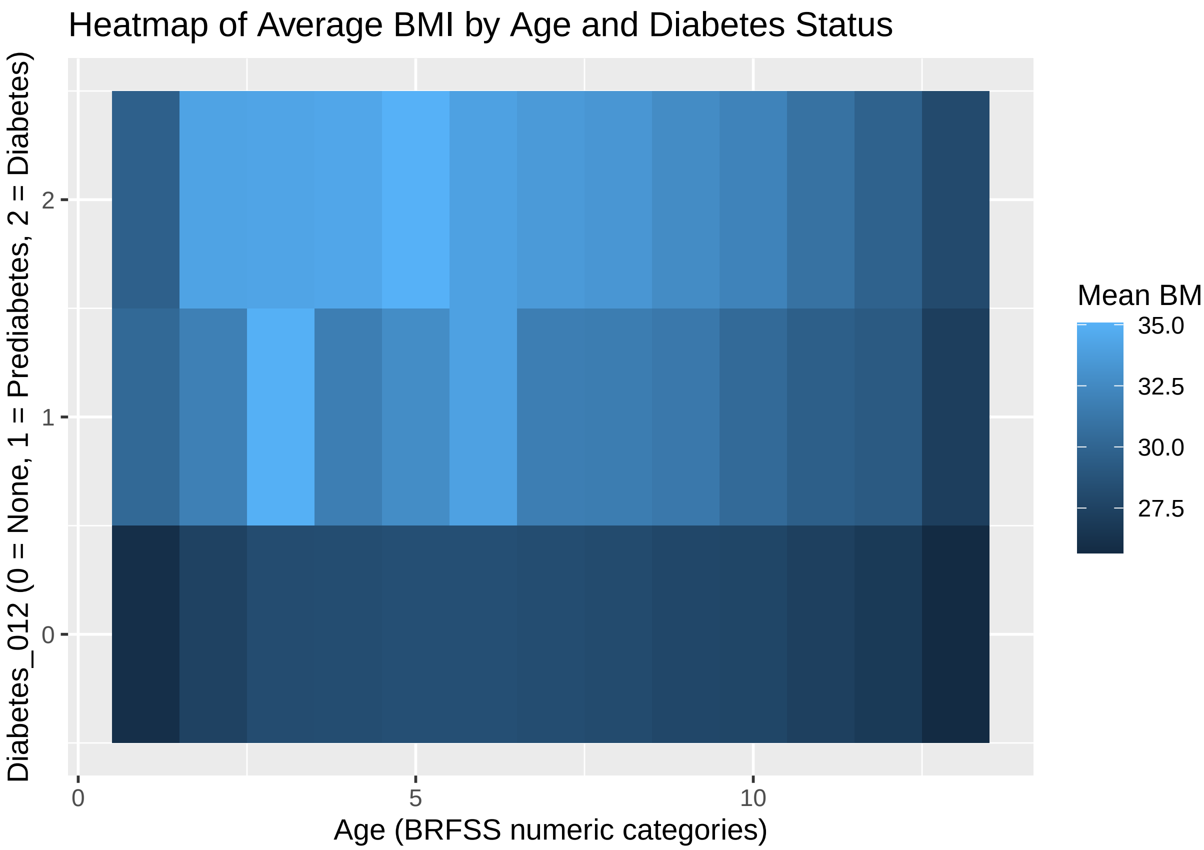


Iris's EDA

```
heat_data <- diabetes_dataset %>%
  group_by(Age, Diabetes_012) %>%
  summarize(mean_BMI = mean(BMI))
```

```
## `summarise()` has grouped output by 'Age'. You can override using the `.groups`
## argument.
```

```
ggplot(heat_data,
       aes(x = Age, y = Diabetes_012, fill = mean_BMI)) +
  geom_tile() +
  labs(
    title = "Heatmap of Average BMI by Age and Diabetes Status",
    x = "Age (BRFSS numeric categories)",
    y = "Diabetes_012 (0 = None, 1 = Prediabetes, 2 = Diabetes)",
    fill = "Mean BMI"
  )
```



Heatmap of Average BMI by Age and Diabetes Status
Diabetes_012 (0 = None, 1 = Prediabetes, 2 = Diabetes)
2
1
0
0
5
10
Age (BRFSS numeric categories)
Mean BMI
35.0
32.5
30.0
27.5